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Abstract

The preparation of a college timetable is one of the most important and challenging tasks in educational institutions. Traditionally, timetables are prepared manually, which requires considerable time and effort while often resulting in scheduling conflicts, duplicate lecture assignments, inefficient classroom utilization, and difficulties in managing faculty workloads. As the number of departments, faculty members, classrooms, and subjects increases, manual timetable preparation becomes increasingly complex and prone to human errors. Therefore, there is a growing need for an automated system that can simplify the timetable generation process while ensuring accuracy and efficient resource utilization.

The **Automated College Timetable Generator** is a web-based application developed to automate the process of timetable generation for educational institutions. The system provides an integrated platform for managing faculty information, subjects, classrooms, faculty preferences, time slots, and timetable generation. It eliminates the need for manual scheduling by automatically allocating subjects to faculty members, classrooms, and lecture slots based on predefined constraints and available resources.

The proposed system has been developed using **HTML5, CSS3, Bootstrap, JavaScript, Node.js, Express.js, and MySQL**. HTML, CSS, and Bootstrap are used to design a responsive and user-friendly interface, while JavaScript provides client-side functionality. Node.js and Express.js are used for developing the backend server and RESTful APIs, and MySQL serves as the relational database management system for storing all academic information securely.

The application consists of several functional modules, including Administrator Login, Dashboard, Faculty Management, Subject Management, Classroom Management, Faculty Preference Management, Time Slot Management, Automatic Timetable Generation, and PDF Export. These modules work together to ensure efficient management of academic scheduling activities. The administrator can enter all required data through an intuitive interface, and the system automatically generates a conflict-free timetable, which can be viewed online or downloaded as a PDF for printing and future reference.

The developed application significantly reduces manual effort, minimizes scheduling conflicts, improves faculty workload distribution, and ensures efficient utilization of classrooms and lecture timings. The modular architecture of the system makes it scalable and easy to maintain. Furthermore, the project provides a strong foundation for future enhancements such as Artificial Intelligence-based timetable optimization, Genetic Algorithm scheduling, student and faculty login modules, attendance management, cloud deployment, mobile application support, and integration with institutional ERP systems.

Extensive testing was conducted to verify the functionality and performance of the application. The results demonstrated that all modules operated successfully and satisfied the specified functional and non-functional requirements. The generated timetable was accurate, reliable, and free from major scheduling conflicts, confirming the effectiveness of the proposed solution.

In conclusion, the **Automated College Timetable Generator** offers a practical, reliable, and efficient solution for academic timetable management. The project demonstrates the effective application of software engineering principles, database management systems, and modern web development technologies in solving a real-world problem faced by educational institutions. The system not only improves administrative efficiency but also provides a scalable platform for future technological advancements in academic scheduling.

Keywords: Automated Timetable Generator, Timetable Management, Node.js, Express.js, MySQL, HTML, CSS, JavaScript, Bootstrap, Web Application, Database Management, Academic Scheduling, REST API.

INTRODUCTION

1.1 Introduction

In today's educational institutions, preparing a college timetable is one of the most important and challenging administrative tasks. A timetable is a schedule that organizes lectures, laboratories, faculty members, classrooms, and time slots in a systematic manner. In many colleges, the timetable is still prepared manually using spreadsheets or paper-based methods. This manual process requires significant time and effort and often results in scheduling conflicts, duplicate lecture assignments, classroom clashes, and inefficient utilization of available resources.

With the rapid growth of Information Technology and web-based applications, educational institutions are increasingly adopting automated systems to improve efficiency and accuracy. An Automated College Timetable Generator provides a computerized solution for generating timetables while considering various constraints such as faculty availability, classroom capacity, subject allocation, lecture hours, and faculty preferences. The system minimizes human intervention, reduces errors, and generates an optimized timetable within a short period of time.

The **Automated College Timetable Generator** developed in this project is a web-based application that simplifies the entire timetable management process. It provides separate modules for managing faculty information, subjects, classrooms, faculty preferences, and time slots. The system automatically generates a timetable by allocating subjects to available faculty members, classrooms, and lecture slots. It also allows administrators to view the generated timetable and download it in PDF format.

The application has been developed using modern web technologies including **HTML5, CSS3, Bootstrap, JavaScript, Node.js, Express.js, and MySQL**. HTML and CSS are used to design the user interface, JavaScript is used for client-side validation and dynamic functionality, Node.js and Express.js are used to develop the backend server and REST APIs, while MySQL is used as the relational database management system for storing all project data.

The system follows a three-tier architecture consisting of the presentation layer (frontend), application layer (backend), and database layer. The frontend communicates with the backend through REST APIs, while the backend performs CRUD (Create, Read, Update, Delete) operations on the MySQL database. This architecture improves scalability, maintainability, and performance.

The proposed system eliminates many limitations of the traditional manual timetable preparation process. It reduces the workload of administrators, avoids timetable conflicts, improves resource utilization, and provides an easy-to-use interface for managing academic schedules. The system is designed to be scalable so that additional features such as Artificial Intelligence-based timetable optimization, student login, faculty login, attendance management, cloud deployment, and mobile application support can be incorporated in future versions.

Overall, the **Automated College Timetable Generator** is an efficient, reliable, and user-friendly solution that enhances the academic scheduling process in educational institutions. It demonstrates the practical implementation of web development, database management, and software engineering concepts to solve a real-world problem faced by colleges and universities.

1.2 Background of the Study

Educational institutions handle a large number of academic activities every semester. One of the most important responsibilities of the administration is to prepare an accurate timetable for all departments. Preparing a timetable manually becomes increasingly difficult as the number of faculty members, subjects, classrooms, and students increases. Any mistake in scheduling may lead to conflicts such as assigning the same classroom to different classes at the same time or allocating multiple lectures to the same faculty member simultaneously.

To overcome these challenges, educational institutions require an automated timetable generation system that can efficiently allocate available resources while minimizing scheduling conflicts. Such a system saves time, reduces administrative workload, and improves the overall academic management process.

1.4 Objectives of the Project

The main objectives of the project are:

- To develop a web-based timetable management system.
- To automate timetable generation.
- To manage faculty, subjects, classrooms, and preferences.
- To reduce manual effort and scheduling conflicts.
- To generate a conflict-free timetable.
- To provide an easy-to-use interface.
- To store academic data securely in a database.
- To generate downloadable PDF reports.

LITERATURE REVIEW

2.1 Introduction

A literature review is a systematic study of previously published research papers, journals, conference proceedings, books, and technical articles related to a particular research topic. It helps researchers understand the existing work, identify research gaps, compare different approaches, and develop improved solutions.

In the field of academic scheduling, timetable generation has always been considered a complex optimization problem because it involves allocating limited resources such as faculty members, classrooms, subjects, and time slots without creating scheduling conflicts. Many researchers have proposed different techniques to automate timetable generation using algorithms, optimization methods, artificial intelligence, and database management systems.

The **Automated College Timetable Generator** presented in this project has been developed after studying several existing timetable management systems and research papers. The literature review helped identify the limitations of current systems and guided the development of a more efficient, user-friendly, and database-driven web application.

2.2 Research Gap

Although several timetable generation systems are available, many problems still exist.

The major research gaps identified are:

- Manual intervention is still required in many systems.
- Faculty preferences are often ignored.
- Classroom allocation conflicts may occur.
- Timetable modification is difficult.
- Limited reporting facilities.
- Poor user interface.
- Lack of PDF export functionality.
- No centralized management of academic data.
- Limited scalability.
- Difficulty in maintaining large institutional data.

These gaps indicate the need for an improved automated timetable generation system.

2.3 Proposed Solution

The proposed **Automated College Timetable Generator** addresses these limitations by providing:

- Web-based application.
- Faculty management.
- Subject management.
- Classroom management.
- Faculty preference management.
- Time slot management.
- Automatic timetable generation.
- PDF download facility.
- Responsive user interface.
- Database-driven architecture.
- REST API-based communication.
- Easy future expansion.

2.4 Advantages of the Proposed System

The proposed system provides several advantages over existing systems.

- Reduces manual work.
- Saves time.
- Generates accurate timetables.
- Avoids scheduling conflicts.
- Improves faculty management.
- Efficient classroom allocation.
- Better resource utilization.
- Easy maintenance.
- Secure database storage.
- Responsive web interface.
- Easy future enhancement.

3.PROPOSED FRAMEWORK

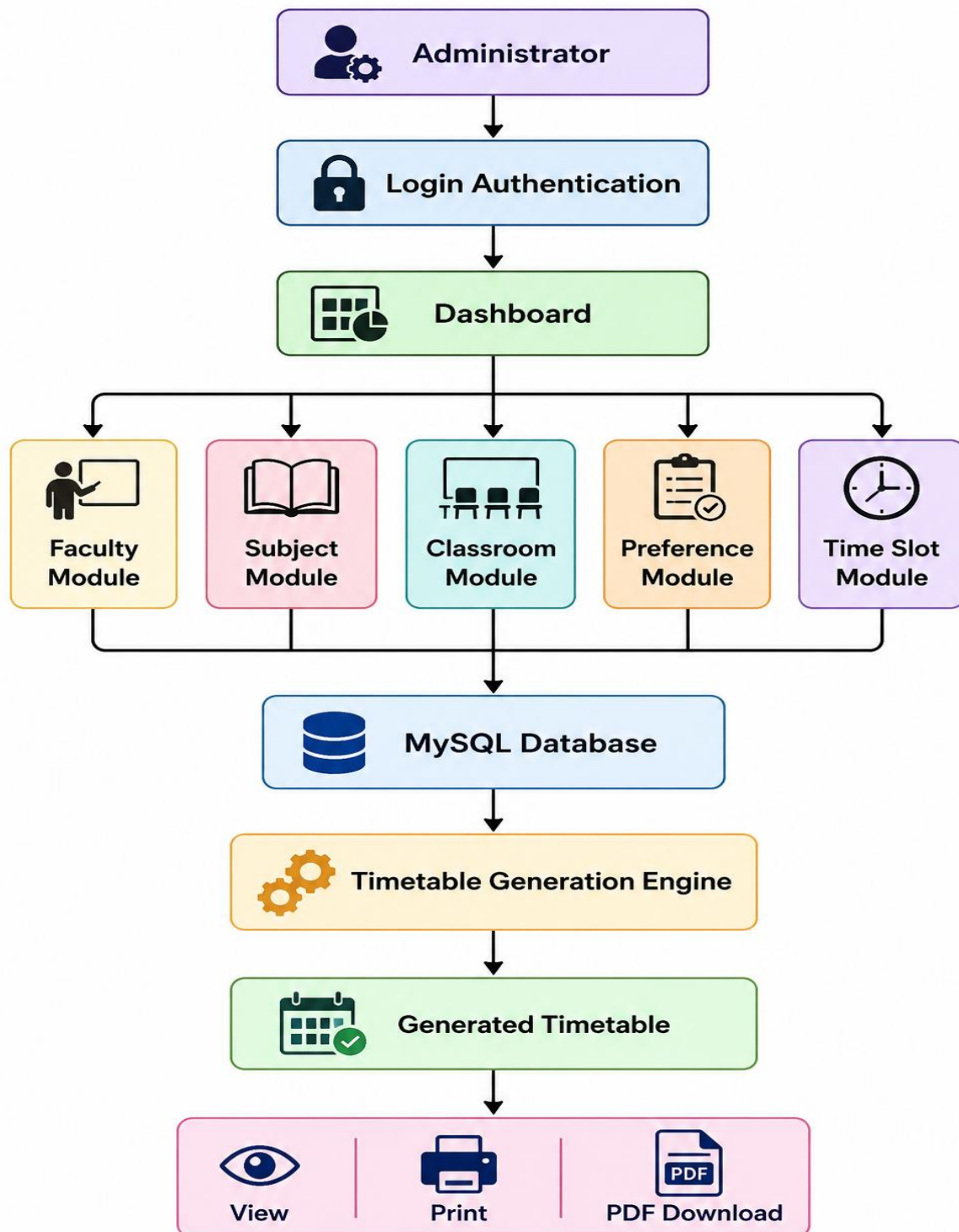
Proposed Framework

The proposed framework of the **Automated College Timetable Generator** is designed to automate the process of timetable preparation by integrating all academic scheduling activities into a single web-based system. The framework follows a **three-tier architecture**, consisting of the Presentation Layer (Frontend), Application Layer (Backend), and Database Layer (MySQL). This architecture ensures efficient communication between the user interface, application logic, and database.

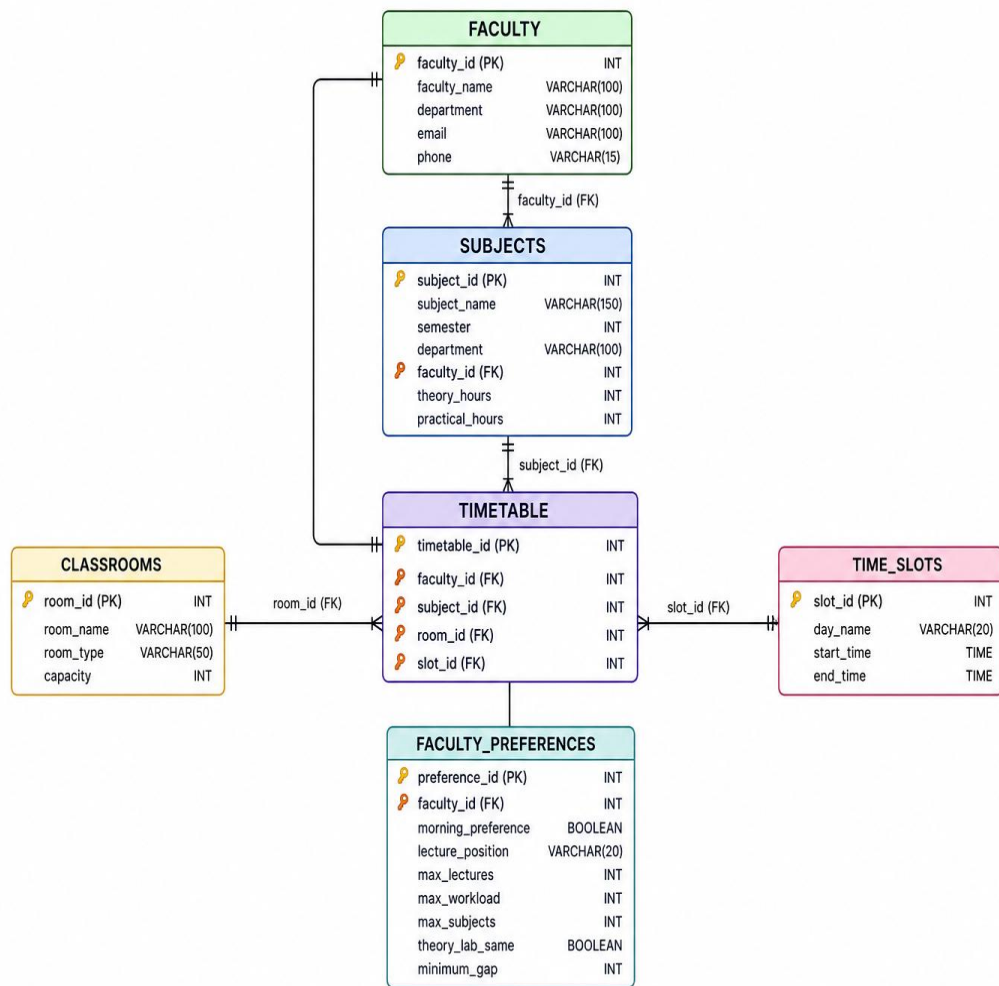
The administrator first logs into the system using valid credentials. After successful authentication, the administrator can access different modules such as Faculty Management, Subject Management, Classroom Management, Faculty Preferences, Time Slot Management, and Timetable Generation. All information entered by the administrator is stored securely in the MySQL database.

When the administrator clicks the **Generate Timetable** button, the backend retrieves faculty details, subject information, classroom data, time slots, and faculty preferences from the database. The timetable generation module processes these records and automatically allocates subjects to available faculty members, classrooms, and lecture slots while avoiding scheduling conflicts. The generated timetable is then stored in the database and displayed to the administrator. The timetable can also be downloaded as a PDF for future reference.

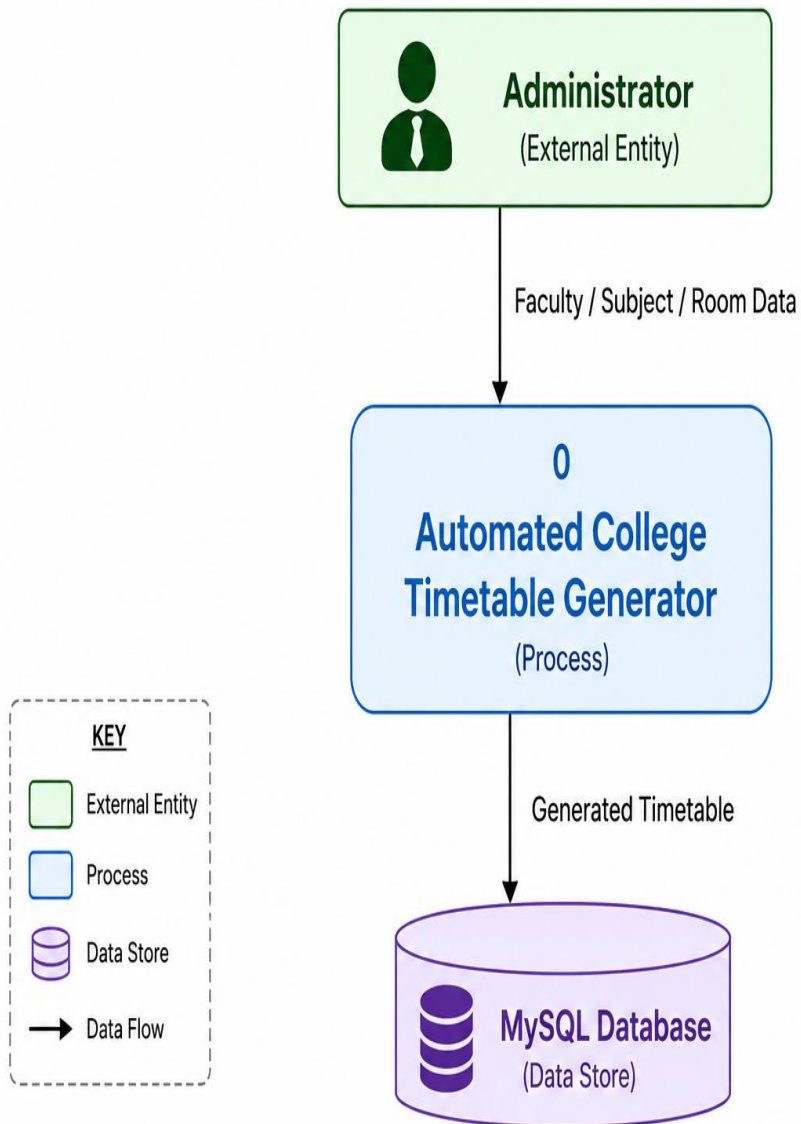
The proposed framework minimizes manual effort, reduces scheduling conflicts, improves resource utilization, and provides a centralized platform for timetable management.



3.2 Proposed Framework Workflow



DFD Level 0 Diagram



3 Proposed Work

The proposed work focuses on developing a complete web-based timetable management system capable of automatically generating college timetables. The system is developed using **HTML, CSS, Bootstrap, JavaScript, Node.js, Express.js, and MySQL.**

The proposed work consists of the following activities:

Administrator Login

The administrator logs into the application using authorized credentials. Only authenticated users can access the system.

Faculty Management

The administrator can add, update, delete, and view faculty information including department, email, and contact details.

Subject Management

The administrator can assign subjects to faculty members along with semester details, theory hours, and practical hours.

Classroom Management

The administrator manages classroom details such as room name, room type, and seating capacity.

Faculty Preference Management

Faculty teaching preferences such as morning preference, lecture position, maximum lectures, workload, and minimum lecture gap are stored.

Time Slot Management

The administrator defines lecture timings for different working days.

Automatic Timetable Generation

The system automatically allocates subjects to available faculty members and classrooms based on the stored information.

Timetable Display

The generated timetable is displayed in tabular format for easy viewing.

PDF Export

The administrator can download the timetable in PDF format for printing and record maintenance.

3.4 Requirement of Hardware and Software Components

Hardware Requirements

The minimum hardware requirements for the successful execution of the Automated College Timetable Generator are as follows.

Hardware Component Specification

Processor	Intel Core i3 (11th Generation) or Higher
RAM	Minimum 8 GB
Storage	256 GB SSD or Higher
Display	1366 × 768 Resolution or Higher
Keyboard	Standard Keyboard
Mouse	Optical Mouse
Internet Connection	Broadband/Wi-Fi

Software Description

HTML5

HTML5 is used for creating the structure of all web pages.

CSS3

CSS3 provides an attractive and responsive user interface.

Bootstrap

Bootstrap simplifies responsive web design using predefined components.

JavaScript

JavaScript provides dynamic functionality, validation, and interaction with the backend.

Node.js

Node.js executes JavaScript on the server side and handles backend processing.

Express.js

Express.js is used for creating REST APIs and handling HTTP requests.

MySQL

MySQL stores faculty records, subjects, classrooms, preferences, time slots, and generated timetables.

Visual Studio Code

Visual Studio Code is used as the Integrated Development Environment (IDE) for coding and debugging.

Google Chrome

Google Chrome is used to execute and test the web application.

Advantages of the Proposed Framework

- Automates timetable generation.
- Reduces manual effort.
- Eliminates scheduling conflicts.
- Improves faculty and classroom utilization.
- Provides centralized data management.
- Generates printable PDF reports.
- Easy to maintain and scalable.
- User-friendly web interface.
- Faster timetable preparation.
- Supports future enhancements such as AI-based scheduling and cloud deployment.

4.Objectives of the Proposed System

The primary objective of the Automated College Timetable Generator is to develop a web-based application that automates the process of generating college timetables efficiently and accurately. The system aims to reduce manual effort, eliminate scheduling conflicts, and provide an easy-to-use platform for managing academic resources such as faculty members, subjects, classrooms, and lecture schedules.

The detailed objectives of the proposed system are as follows:

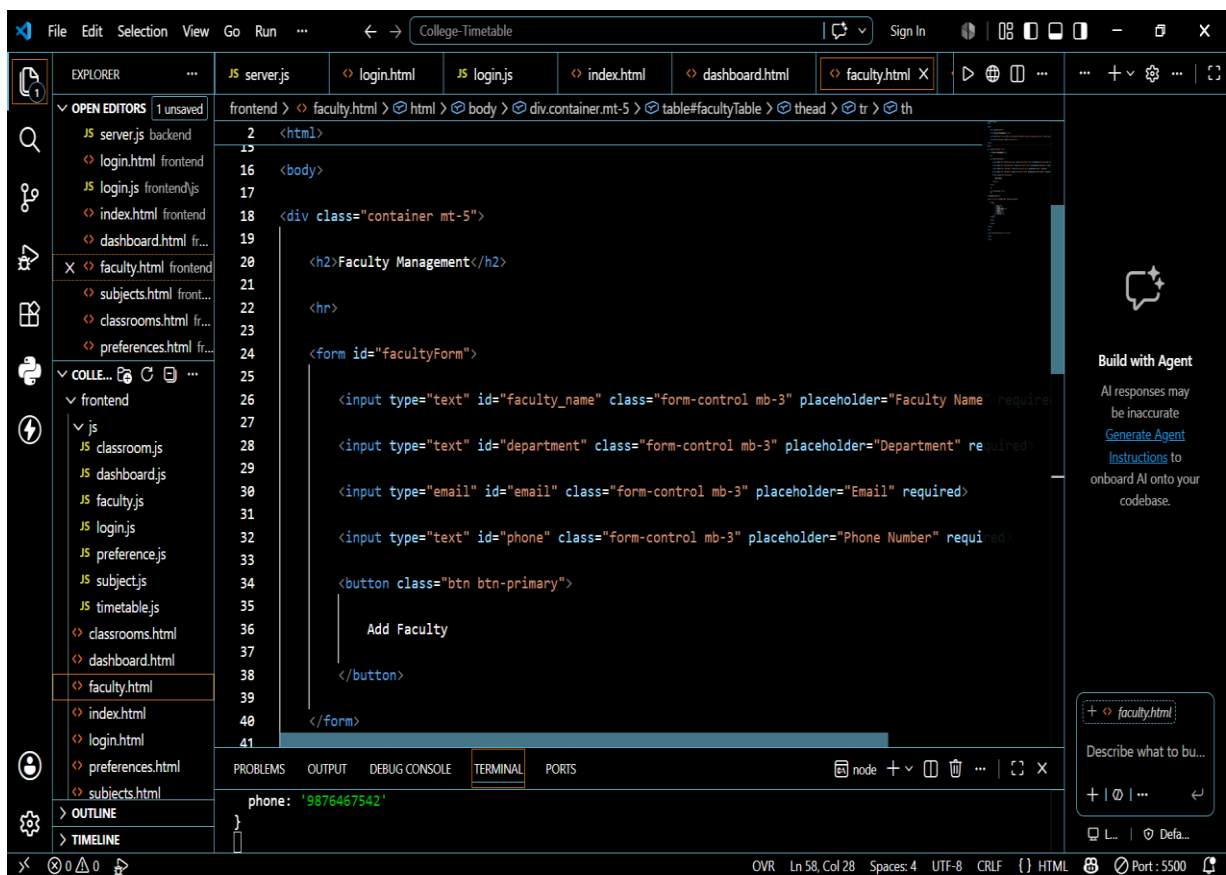
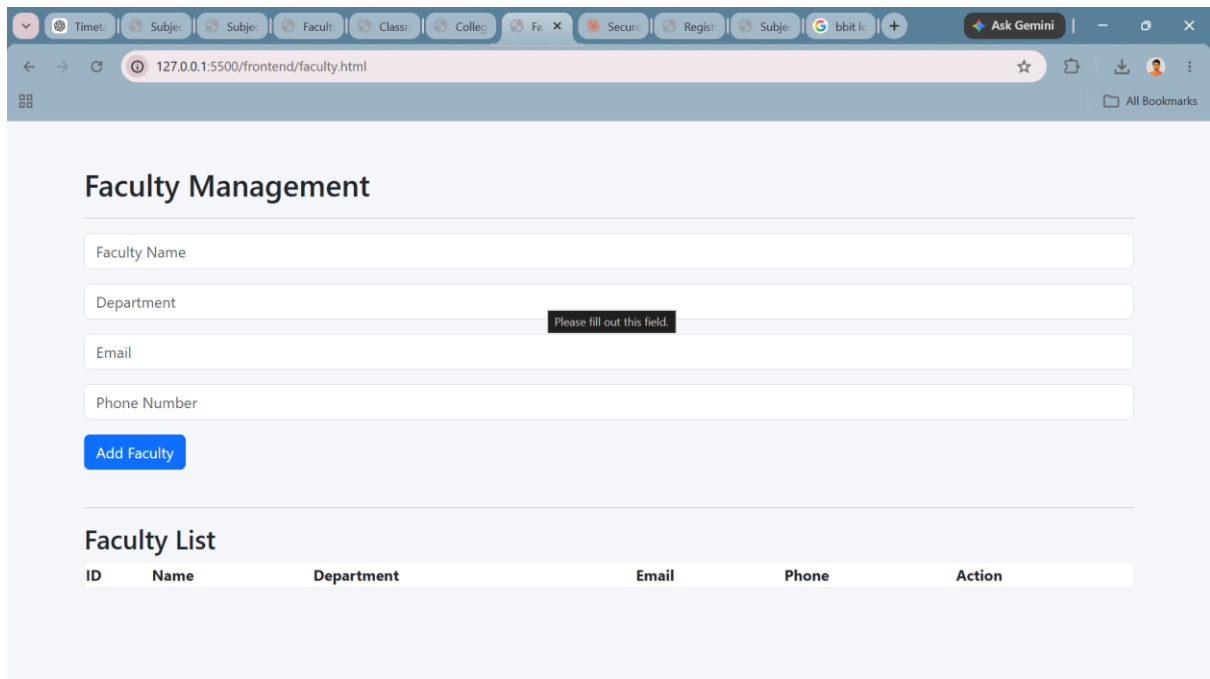
Primary Objectives

- To develop an automated timetable generation system for educational institutions.
- To eliminate the manual process of timetable preparation.
- To reduce scheduling conflicts among faculty members, classrooms, and lecture timings.
- To improve the overall efficiency of timetable management.
- To provide a centralized database for storing academic information.

Specific Objectives

1. Faculty Management

- To maintain faculty information in a centralized database.
- To add, update, delete, and view faculty records.
- To manage faculty departments and contact details.



2. Subject Management

- To maintain complete information about subjects.
- To assign subjects to faculty members.
- To manage theory and practical lecture hours.

The image shows a development environment with VS Code and a web browser. The VS Code editor displays the 'subjects.html' file, which contains HTML code for a subject management form. The code includes a form with fields for Subject Name, Semester, Department, Faculty (a select dropdown), Theory Hours, and Practical Hours, along with an 'Add Subject' button. The web browser shows the rendered form at the URL '127.0.0.1:5500/frontend/subjects.html'.

```
front-end > subjects.html > html > body > div.container.mt-5 > form#subjectForm > div.mb-3 > option
2  <html lang="en">
16 <body>
18 <div class="container mt-5">
24 <form id="subjectForm">
38 <div class="mb-3">
44 <div class="mb-3">
45 <label>Faculty</label>
46
47 <select id="faculty_id" class="form-select" required>
48 |   <option value="">Select Faculty</option>
49 </select>
50
51 </div>
52
53 <option value="">
54 Select Faculty
55 </option>
56
57 </select>
58
59 </div>
60
61 <div class="mb-3">
62 <input type="number" id="theory_hours" class="form-control" placeholder="Theory Hours" required>
63 </div>
64
65 <div class="mb-3">
```

Build with Agent
AI responses may be inaccurate
[Generate Agent](#)
Instructions to onboard AI onto your codebase.

Describe what to bu...
+ | | ...

127.0.0.1:5500/frontend/subjects.html

Subjects Management

Subject Name

Semester

Department

Faculty

Faculty

Select Faculty

Select Faculty

Theory Hours

Practical Hours

Add Subject

3. Classroom Management

- To maintain classroom details.
- To store room type and seating capacity.
- To allocate classrooms efficiently during timetable generation.

The screenshot displays the development environment for the 'College-Timetable' application. The VS Code editor shows the `classrooms.html` file in the `frontend` directory. The code defines an HTML form for adding a classroom with fields for Room Name, Room Type, and Capacity. The terminal shows the output of a Node.js script, displaying a phone number: '9876467542'.

The web browser shows the rendered application. The 'Classroom Management' section includes a form with the following fields:

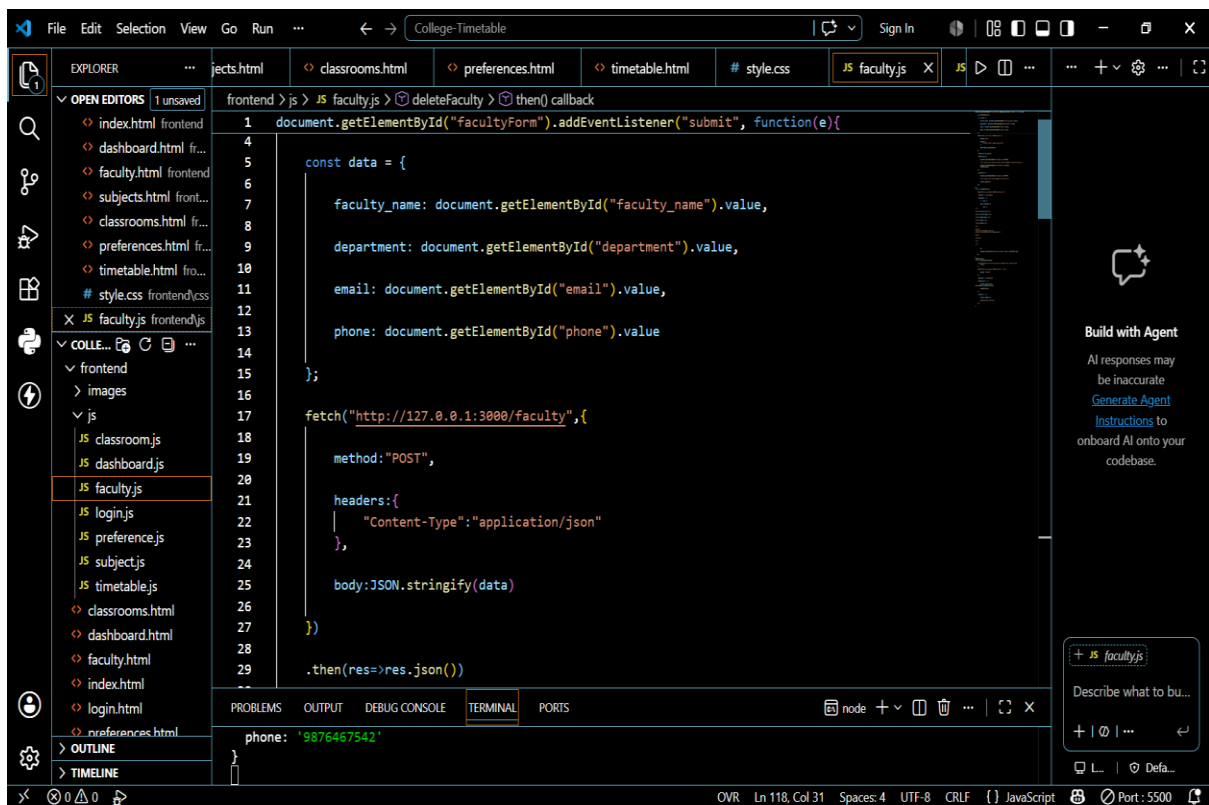
- Room Name
- Theory
- Capacity

The 'Add Classroom' button is green. Below the form is the 'Classroom List' table:

ID	Room Name	Room Type	Capacity
1	A101	Theory	60
2	Lab-1	Lab	30
3	A101	Theory	50

4. Faculty Preference Management

- To store faculty teaching preferences.
- To manage maximum lecture workload.
- To define lecture position preferences.
- To maintain minimum lecture gaps.

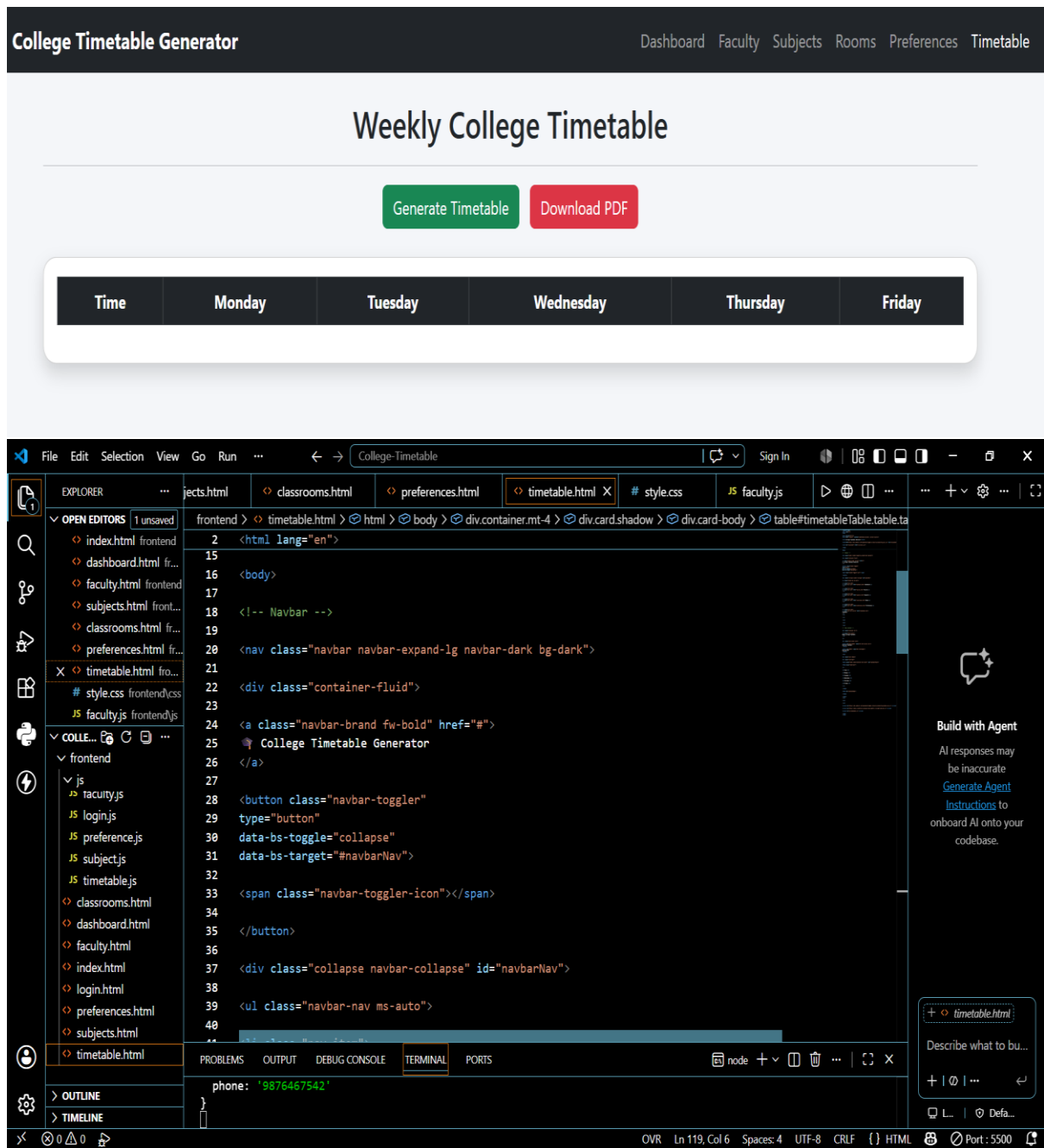


The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays the project structure for 'College-Timetable', including folders for 'frontend' and 'js', and various HTML and JavaScript files. The 'faculty.js' file is selected and open in the main editor. The code in 'faculty.js' defines a JavaScript object 'data' containing faculty details and uses the 'fetch' API to send a POST request to a server endpoint. The terminal at the bottom shows the output of the code execution, displaying the phone number '9876467542'.

```
frontent > js > JS faculty.js > deleteFaculty > then() callback
1 document.getElementById("facultyForm").addEventListener("submit", function(e){
4
5     const data = {
6
7         faculty_name: document.getElementById("faculty_name").value,
8
9         department: document.getElementById("department").value,
10
11         email: document.getElementById("email").value,
12
13         phone: document.getElementById("phone").value
14     };
15
16     fetch("http://127.0.0.1:3000/faculty",{
17
18         method:"POST",
19
20         headers:{
21             "Content-Type":"application/json"
22         },
23
24         body:JSON.stringify(data)
25     })
26
27     .then(res=>res.json())
28
29     phone: '9876467542'
```

5. Time Slot Management

- To manage lecture timings for all working days.
- To maintain start and end times of each lecture.



6. Automatic Timetable Generation

- To generate conflict-free timetables automatically.
- To allocate faculty members based on subject assignments.
- To assign classrooms according to availability.
- To optimize resource utilization.

7. Timetable Display

- To display the generated timetable in a user-friendly tabular format.
- To allow administrators to verify the generated schedule.

8. PDF Generation

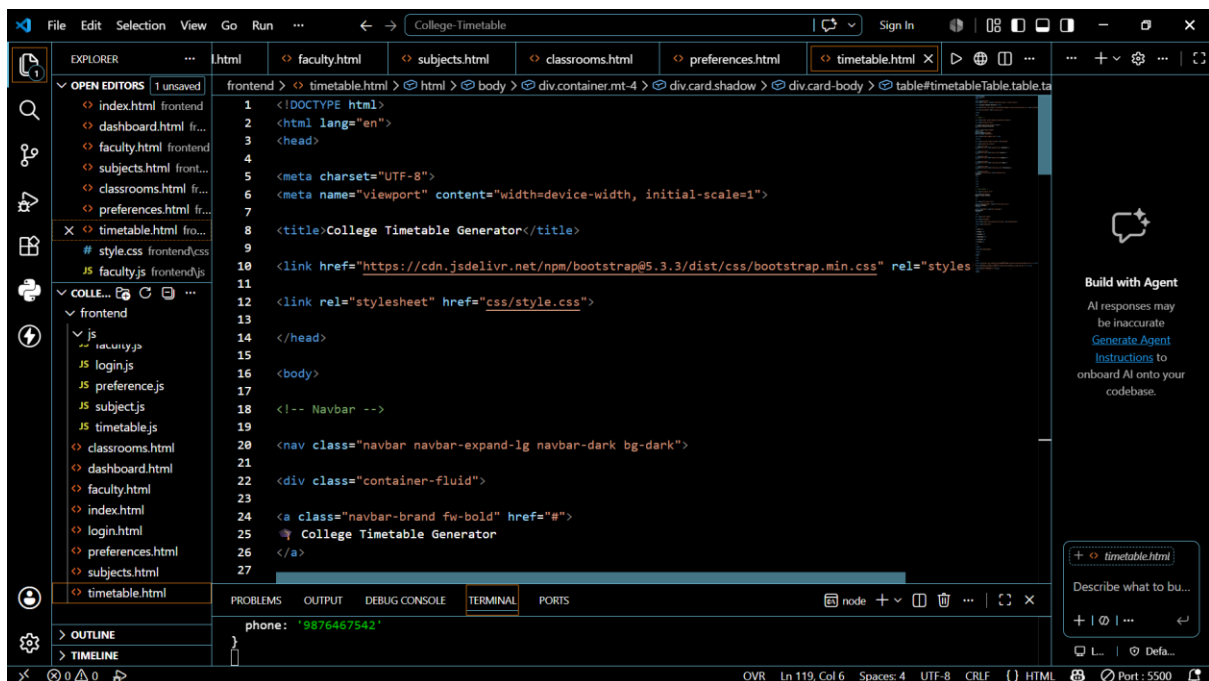
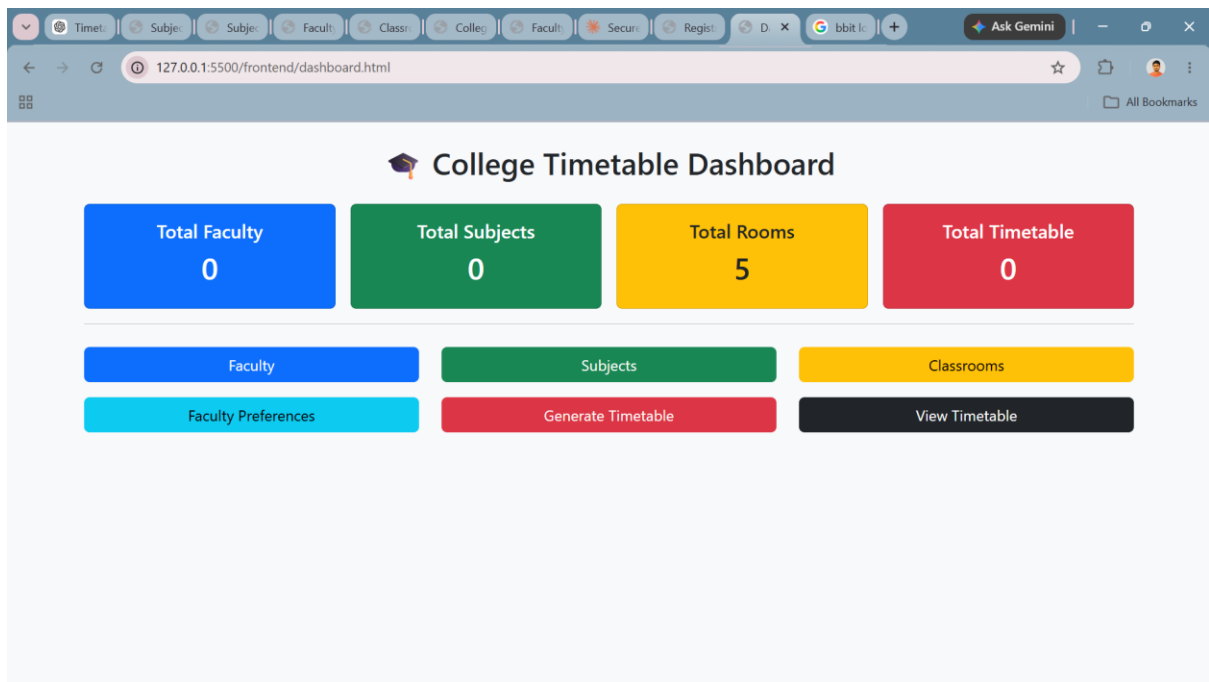
- To provide a downloadable PDF version of the timetable.
- To facilitate printing and record maintenance.

9. Database Management

- To securely store all project data in MySQL.
- To maintain data consistency using primary and foreign keys.

10. User-Friendly Interface

- To provide an attractive, responsive, and easy-to-use web interface.
- To simplify system navigation for administrators.



5. CONCLUSION

The **Automated College Timetable Generator** was successfully designed, developed, and implemented to automate the process of timetable preparation in educational institutions. The project effectively addresses the limitations of traditional manual timetable scheduling by providing a fast, accurate, and user-friendly web-based solution. The system integrates all essential academic scheduling activities, including faculty management, subject management, classroom management, faculty preference management, time slot management, automatic timetable generation, and PDF report generation.

The proposed system significantly reduces the time and effort required for timetable preparation while minimizing scheduling conflicts such as faculty clashes, classroom allocation conflicts, and duplicate lecture assignments. By storing all information in a centralized MySQL database, the application ensures secure data management, consistency, and easy retrieval of records. The use of modern web technologies such as **HTML5, CSS3, Bootstrap, JavaScript, Node.js, Express.js, and MySQL** makes the application reliable, scalable, and easy to maintain.

The project follows a modular architecture, allowing each module to function independently while maintaining smooth integration with other components. The frontend provides an attractive and responsive user interface, whereas the backend efficiently handles data processing and communication with the database through REST APIs. The generated timetable can be viewed on the system and exported as a PDF, making it convenient for administrators to print and distribute academic schedules.

Extensive testing confirmed that all modules of the application function correctly according to the specified requirements. The Faculty Module, Subject Module, Classroom Module, Faculty Preference Module, and Timetable Generation Module successfully performed their intended operations without major errors. The generated timetable was accurate, well-organized, and stored correctly in the database.

The implementation of the Automated College Timetable Generator demonstrates the practical application of software engineering principles, database management systems, and web development technologies in solving

a real-world academic scheduling problem. The project not only improves administrative efficiency but also provides a strong foundation for future enhancements and large-scale deployment in educational institutions.

Overall, the proposed system achieves its primary objective of developing an efficient, reliable, and automated timetable management solution. It reduces manual work, improves resource utilization, enhances scheduling accuracy, and provides a modern platform for managing academic timetables. Therefore, the **Automated College Timetable Generator** can be considered a successful and effective solution for timetable management in colleges and universities.

Future Scope

The **Automated College Timetable Generator** has been successfully developed to automate the timetable generation process in educational institutions.

Although the current system fulfills its primary objectives, there are several opportunities for future enhancements that can improve its efficiency, intelligence, usability, and scalability. Future developments can transform the application into a comprehensive academic management system capable of handling the requirements of large universities and multi-campus institutions.

The following enhancements are proposed for future versions of the system:

1. Artificial Intelligence (AI) Based Timetable Generation

The current system generates timetables based on predefined rules and available data. In future, Artificial Intelligence techniques can be incorporated to generate smarter and more optimized timetables by automatically analyzing faculty availability, classroom utilization, workload distribution, and institutional constraints. AI can significantly reduce scheduling conflicts and improve resource allocation.

2. Genetic Algorithm Optimization

Genetic Algorithms (GA) are widely used for solving optimization problems. The timetable generation process can be enhanced by implementing a Genetic Algorithm to produce highly optimized timetables that satisfy multiple constraints such as faculty preferences, classroom availability, and subject priorities simultaneously.

3 Mobile Application Development

A mobile application for Android and iOS platforms can be developed to provide easy access to timetables. Faculty members and students can receive real-time notifications regarding timetable changes through the mobile application.

4 Multi-Department and Multi-Campus Support

The current application is designed for a single department. Future versions can support multiple departments and campuses within the same institution. This enhancement will enable centralized timetable management for large universities.

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- [MDN Web Docs](#)
- [W3Schools](#)
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APPENDIX

Appendix A – Source Code

The source code of the Automated College Timetable Generator has been developed using HTML, CSS, Bootstrap, JavaScript, Node.js, Express.js, and MySQL. The application consists of multiple modules, each performing a specific function.

Major Source Files

Frontend

- login.html
- dashboard.html
- faculty.html
- subjects.html
- classrooms.html
- preferences.html
- timetable.html

JavaScript Files

- login.js
- dashboard.js
- faculty.js
- subject.js
- classroom.js
- preferences.js
- timetable.js

CSS

- style.css

Backend

- server.js
- package.json

Appendix C – REST APIs

API Endpoint	Method	Description
/faculty	POST	Add Faculty
/faculty-list	GET	View Faculty
/faculty/:id	DELETE	Delete Faculty
/subjects	POST	Add Subject
/subjects-list	GET	View Subjects
/classrooms	POST	Add Classroom
/classrooms-list	GET	View Classrooms
/faculty-preferences	POST	Save Faculty Preference
/faculty-preferences-list	GET	View Preferences
/generate-timetable	POST	Generate Timetable
/timetable	GET	View Timetable